

Lab Assignment 1

Topic: Recalling Python

Q1. Write a python code to roll a two dice and the output should be as follows:

```
Roll the dice? (y/n): y
(5, 6)
Roll the dice? (y/n): h
Invalid choice!
Roll the dice? (y/n): n
Thanks for playing!
```

Q2. Write a python code to guess the number generated by computer between 0 to 100 and output should be as follows:

```
Enter any number between 0 to 100: 50
The number you have entered is too low
Enter any number between 0 to 100: 111
Enter the valid number
Enter any number between 0 to 100: 70
The number you have entered is too low
Enter any number between 0 to 100: 90
The number you have entered is too high
Enter any number between 0 to 100: 80
The number you have entered is too low
Enter any number between 0 to 100: 85
The number you have entered is too low
Enter any number between 0 to 100: 88
The number you have entered is too low
Enter any number between 0 to 100: 89
Generated number is same as your input: 89
```

Q3. Write a python code for Rock, Paper and Scissor game and output should be as follows:

```
Enter your choice (r,p,s): r
You have chosen 🪨
Computer has chosen 🪨
Game is tie
Do you want to play again? (y/n) : y
Enter your choice (r,p,s): s
You have chosen ✂️
Computer has chosen ✂️
Game is tie
Do you want to play again? (y/n) : y
Enter your choice (r,p,s): s
You have chosen ✂️
Computer has chosen 🪨
You Lose
Do you want to play again? (y/n) : n
Thank you!
```

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Q4. Check the strength of the password by the following instructions:

Give one point if password contains

- 8 characters' long
- Uppercase letters (A-Z)
- Lowercase letters (a-z)
- Numbers (0-9)
- Special characters (e.g., @, #, \$, %, &)

Then classify it by their points (5: strong, 4: Moderate, 3: less protected, 1 & 2: very poor).

Q5. Write a python program to compute Mean, Median, Mode, and Variance without using any library or inbuilt functions.

Q6. Study of the NumPy Library